

Chapter 9 Relations

Chapter Summary

- Relations and their Properties
- n -ary Relations and their Applications
- Representing Relations
- Closures of Relations
- Equivalence Relations
- Partial Orderings

Section 9.1 Relations and Their Properties

Section Summary

- Relations and Functions
- Properties of Relations
 - Reflexive Relations
 - Symmetric and Antisymmetric Relations
 - Transitive Relations
- Combining Relations

Binary Relations

Definition:

A **binary relation** R from A to B is a subset $R \subseteq A \times B$.

Relation is more general than functions. For functions, elements in A all have corresponding element in B .

Binary Relation on a Set

Definition

A binary relation R on A is a subset of $A \times A$ or a relation from A to A .

Reflexive Relations

Definition:

R is reflexive if and only if

$$\forall x \in A \quad (x, x) \in R$$

Symmetric Relations

Definition:

R is symmetric if and only if

$$xy(x, y) \ R \ (y, x) \ R$$

Antisymmetric Relations

Definition

R is antisymmetric if and only if

$$xy(x, y) \ R \wedge (y, x) \ R \ x = y$$

Transitive Relations

Definition:

R is transitive if and only if

$$xyz(x, y) \ R \wedge (y, z) \ R \ (x, z) \ R$$

For above, note for the situations that **the premise is false** then the relation is true.

Combining Relations

The combination of relations: $R_1 \cap R_2, R_1 \cup R_2, R_1 - R_2, R_2 - R_1$.

Composition

Definition:

$R_1 : A \rightarrow B, R_2 : B \rightarrow C$, then the composition of R_2 with R_1 is $R_2 \circ R_1 : A \rightarrow C$ where if $(x, y) \in R_1, (y, z) \in R_2$, then $(x, z) \in R_2 \circ R_1$.

Powers of a Relation

Definition

$R : A \rightarrow A$ and R is a binary relation, then the power R^n is defined inductively by

- Basic Step: $R_1 = R$
- Inductive Step: $R^{n+1} = R^n \circ R$.

Theorem 1:

The relation R on a set A is transitive if and only if

$$R^n \subseteq R$$

for $n = 1, 2, \dots$.

(proved by mathematical induction)

Section 9.3 Representing Relations

Section Summary

- Representing Relations using Matrices
- Representing Relations using Digraphs

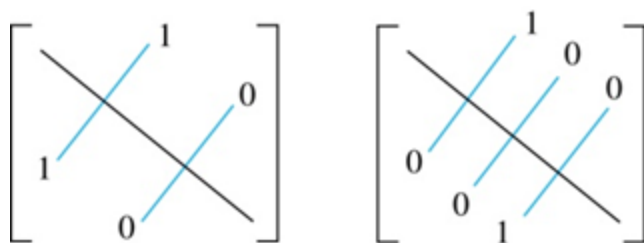
Representing Relations Using Matrices

Suppose R is relation from $A = \{a_1, a_2, \dots, a_m\}$ to $B = \{b_1, b_2, \dots, b_n\}$, then R can be represented by matrix $M_R = m_{ij}$ where

$$m_{ij} = \begin{cases} 1 & \text{if } (a_i, b_j) \in R \\ 0 & \text{if } (a_i, b_j) \notin R \end{cases}$$

Matrices of Relations on Sets

- If R is a reflexive relation, all elements on the main diagonal of M_R are equals to 1.
- If R is symmetric relation, $m_{ij} = 1$ whenever $m_{ji} = 1$. If R is antisymmetric relation, $m_{ij} = 0$ or $m_{ji} = 0$ when $i \neq j$.



(a) Symmetric

(b) Antisymmetric

Representing Relations Using Digraphs

Definition:

Vertices set represents elements in the sets, edges set represent the element in the relation set.

Determining which Properties a Relation has from its Digraph

- Reflexivity: A self-loop must be present at all vertices in the graph.
- Symmetry: If (x, y) is an edge then so is (y, x) .
- Antisymmetry: If (x, y) with $x \neq y$ is an edge, then (y, x) is not an edge.
- Transitivity: If $(x, y), (y, z)$ are edges, then so is (x, z) .

- **Powers of Relation**: If there is a path of length n from x to y in R , then (x, y) is in R^n .

Section 9.5 Equivalence Relations

Section Summary

- Equivalent Relations
- Equivalent Classes
- Equivalence Classes and Partitions

Equivalence Relations

Definition 1:

A relation on a set A is **equivalence relation** if it is **reflexive, symmetric and transitive**.

Definition 2:

Two elements a and b are related by equivalence relation are called **equivalent**, denoted by $a \sim b$.

Examples:

- Strings: aRb if and only if $l(a) = l(b)$.
- Congruence Modulo m : $R = \{(a, b) | a \equiv b \pmod{m}\}$.
- Not Equivalent relations: divides relation on the set of positive integers.(not symmetric)

Equivalence Classes

Definition 3:

R is an equivalence relation on A . The **set of all elements that are related to an element a** of A is called the **equivalence class of a** , denoted by a_R .

If $b \in a_R$, then b is called a **representative** of this equivalence class.

The equivalence classes of relation congruence modulo m are called the **congruence classes modulo m** . The congruence class of an integer a modulo m is denoted by a_m thus

$$a_m = \{\dots, a - 2m, a - m, a, a + m, \dots\}.$$

Equivalence Classes and Partitions

Theorem 1:

R is an equivalence relation on a set A . These statements for elements a and b of A are equivalent.

- aRb
- $a \in b_R$

- $a \cap b \neq$

Partition of a Set

Definition:

A partition of S is collection of disjoint nonempty subsets of S that have S as their union. i.e., for collection of subsets A_i ,

- $A_i \neq$
- $A_i \cap A_j = \emptyset \quad i \neq j$
- $\bigcup_i A_i = S$



An Equivalence Relation Partitions a Set

R is equivalence relation on A , then

$$\{a_R\}_{a \in A} = A$$

From theorem 1, $a_R = b_R$ or $a_R \cap b_R = \emptyset$. Therefore, the equivalence classes form a partition of A .

Theorem 2:

R is equivalence relation on S , then the equivalence classes of R form a partition of S .

Conversely, given a partition of S , there is an equivalence relation R that has the partition as its equivalence classes.

Section 9.6 Partial Orderings

Section Summary

- Partial Orderings and Partially-ordered Sets
- Lexicographic Orderings
- Hasse Diagrams
- Lattices

- Topological Sorting

Partial Orderings

Definition 1:

A relation R on a set S is called **partial ordering** if it is **reflexive, antisymmetric and transitive**. A set with partial ordering R is called **partially ordered set** or **poset**, denoted by (S, R) . Members of S are called **elements** of the poset.

Examples:

- $\geq$ relation is a partial ordering on the set of integers.
- Divisibility relation is a partial ordering on the set of integers.
- $\subseteq$ is a partial ordering on the power set of S . (the set of all subset of S)

Comparability

Definition 2:

The elements a and b of a poset $(S, \preceq)$ are **comparable** if either $a \preceq b$ or $b \preceq a$. If neither $a \preceq b$ nor $b \preceq a$, then a and b are called **incomparable**. ($\preceq$ represents relations of any posets)

Definition 3:

If $(S, \preceq)$ is a poset and every two elements are comparable, S is called a **totally ordered** or **linearly ordered** set or a **chain**. And $\preceq$ is called a **total order** or a **linear order**.

Example, $(\mathbb{Z}, |)$

Definition 4:

$(S, \preceq)$ is **well-ordered** set if it is a poset such that $\preceq$ is total ordering and **every nonempty subset of S has a least element**.

Example, $(\mathbb{N}, \leq)$

Lexicographic Order

Definition:

Given two posets $(A_1, \preceq_1)$ and $(A_2, \preceq_2)$, then the **lexicographic ordering** on $A_1 \times A_2$ is defined by

$$(a_1, a_2) \preceq (b_1, b_2) \iff a_1 \preceq_1 b_1 \text{ or } (a_1 = b_1 \text{ and } a_2 \preceq_2 b_2)$$

Hasse Diagrams



Definition:

Hasse Diagram is visual representation of partial ordering that leaves out edges that must be present because of **reflexive and transitive** properties. (Upper c is a Hasse diagram)

Procedure for Constructing a Hasse Diagram

For a poset $(S, \leq)$, start with the directed graph of the relation

- Remove the loops (a, a) at every vertex
- Remove all edges (x, y) where $x \leq z$ and $z \leq y$.
- Arrange each edge so that its initial vertex is below the terminal vertex. Remove all the arrows